

Evaluating Spatio-Temporal Deep Learning Models for Tropical Deforestation Prediction

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Abstract

Tropical forests are important global carbon sinks, but they are under pressure from anthropogenic and environmental deforestation drivers. Current remote sensing systems allow for large-scale monitoring, but they are inherently reactive. They are only capable of registering deforestation events after they have already occurred. Preventing deforestation requires accurate prediction models. Current models aggregate temporal information into a single feature, but explicit spatio-temporal dynamics of deforestation are underexplored in current deep learning algorithms. This thesis investigated the added value of explicitly modeling the temporal dependencies for near-future deforestation prediction. A 2D ResUNet baseline was compared against three spatio-temporal 3D models: a 3D ResUNet, a 3D ConvLSTM and a Video Vision Transformer (ViViT). The main evaluation metric was the $F_{0.5}$ score, and multiple temporal context lengths, prediction horizons and regions (Gabon, Papua New Guinea and Venezuela) were investigated. The results indicated that explicit modeling of spatio-temporal dependencies does not result in a universal, significant improvement over the 2D baseline. The convolutional models significantly outperformed the ViViT, and the 2D ResUNet achieved the highest overall $F_{0.5}$ score. The evaluation of the temporal context showed that a short historical context length of 3 months was sufficient for capturing deforestation dynamics. The 3D ResUNet demonstrated a significant improvement in detecting new deforestation locations in Gabon, but this performance did not generalize to the other tested regions. The findings suggest that for regional datasets, the local focus of convolutional networks and short-term information outperform explicit temporal sequences. It is recommended that transitioning to pan-tropical datasets over longer periods of time could improve the performance of the 3D spatio-temporal models, and that switching to a regression task could more accurately predict the deforestation intensity.

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Acronyms

BCE	Binary Cross-Entropy.
CMIP6	Coupled Model Intercomparison Project Phase 6.
CNN	Convolutional Neural Network.
EUDR	EU Regulation on Deforestation Free products.
FN	False Negatives.
FP	False Positives.
GFW	Global Forest Watch.
GLAD	Global Land Analysis and Discovery.
IDA	Integrated Deforestation Alert.
LSTM	Long Short-term Memory.
MLP	Multilayer Perceptron.
MSE	Mean Squared Error.
PNG	Papua New Guinea.
RAdam	Rectified Adam.
RADD	Radar for Detecting Deforestation.
REDD+	Reducing Emissions from Deforestation and forest Degradation.
RF	Random Forest.
RNN	Recurrent Neural Network.
SVM	Support Vector Machine.
TN	True Negatives.
TP	True Positives.
ViViT	Video Vision Transformer.
WWF	World Wide Fund for Nature.

1 Introduction

Tropical forests play an important role in regulating the global climate and maintaining biodiversity. They function as one of the world's biggest carbon sinks by storing large amounts of carbon in their biomass, which helps with mitigating the effects of anthropogenic CO₂ emissions (Pan et al., 2024). Despite the positive effects that tropical forests have on the environment, they are under pressure from deforestation driven by anthropogenic factors, such as agricultural expansion, settlement expansion and infrastructure development, as well as environmental factors such as forest fires and landslides (Jayathilake et al., 2021). Next to reducing the forest's capacity to act as a carbon sink, deforestation also contributes to greenhouse gas emissions and the loss of biodiversity (Betts et al., 2017; Gatti et al., 2021).

Governments are making attempts to limit deforestation, but despite these attempts, it continues to be a threat to tropical forests (Cremer et al., 2020; Pendrill et al., 2022). Many tropical forests are vast and remote areas where enforcement and monitoring capabilities on the ground are limited. To combat this, multiple global and regional initiatives and approaches have been introduced. Projects such as REDD+ allow users to have near-real time insights into deforestation events in tropical forests, alongside other frameworks and initiatives such as the EU Regulation on Deforestation Free products (EUDR) and targets discussed at COP28 (Iyer et al., 2023; Muradian et al., 2025).

To support this need for monitoring, technical solutions and developments in earth observation can be used. Satellite images from, for example, Landsat and Sentinel-2 provide high resolution spatial data on land coverage, and comparing images from different time periods can give valuable insights into the dynamics of land use change. The data from satellites is used in large-scale monitoring systems such as the Global Forest Watch (GFW) alerts. This system combines three alert types to detect and track forest disturbances (Reiche et al., 2024). It uses GLAD (Global Land Analysis & Discovery) alerts, which uses 30-meter resolution optical images from Landsat; GLAD-S2, which uses a higher resolution of 10 meters of optical Sentinel-2 data; and RADD (Radar for Detecting Deforestation) alerts, which use Sentinel-1 radar data to detect changes through the cloud cover.

While these remote sensing systems are very effective at large scale monitoring and detection, they are only reactive. The systems can only identify forest disturbance events after they have already occurred, which makes them unsuitable for proactive prevention of deforestation (Overlan et al., 2025). To prevent further deforestation, it is important for governments and organizations to have timely and accurate information on where deforestation is likely to take place so that they can use their resources effectively (Jaffé et al., 2021). Deforestation has multiple drivers, such as environmental and human factors, and the progression of deforestation over time is dynamic. Therefore, to be able to prevent forest loss, a good understanding of the spatio-temporal characteristics and dynamics of deforestation is needed.

To accurately predict these events, it is important to understand how deforestation actually progresses in the real world. Spatially speaking, deforestation is a very localized process. It usually expands gradually outward from existing forest edges, new infrastructure or nearby agricultural patches. Temporally, the progression of deforestation is very dynamic and is influenced by multiple factors, for example seasonality (Staal et al., 2020). Deforestation rates can fluctuate throughout the year, depending on these factors. By capturing both the spatial proximity of previous deforestation events, and also the evolving temporal patterns over time, it is expected that more accurate predictions can be made.

Multiple modeling approaches have been developed to anticipate the risk of deforestation, ranging from simple rule-based models to more complex machine learning and deep learning models. Machine learning methods such as Random Forest or Neural Networks have been applied with potential improvements against rule-based models when specific scenario and design decisions are made (Mayfield et al., 2020). More recently, deep learning methods have outperformed traditional models by capturing the complex spatial and temporal patterns of deforestation, for example through Convolutional Neural Networks (CNN) (Ball et al., 2022; Cué La Rosa et al., 2025). Furthermore, recent large-scale benchmarking efforts such as the ForestCast project have shown that deep learning architectures significantly outperform classical machine learning algorithms in predicting deforestation risk (Overlan et al., 2025).

Cué La Rosa et al., 2025 uses a deep learning framework within the Forest Foresight project, on which this paper will further build. The Forest Foresight early warning model uses a Re-sUNet model (Diakogiannis et al., 2020) to predict near-future tropical deforestation locations. Combining spatial and temporal data with other datasets such as topography, information on roads, and protected areas leads to a good foundation for predictive modeling of deforestation (Kaslimi et al., 2023).

Despite these advancements, many current deforestation prediction studies rely heavily on static or temporally aggregated input features rather than explicitly modeling temporal sequences. Although recent approaches such as the Forest Foresight model include temporal information through aggregated features (for example accumulated forest disturbance over the past months), they operate on a single timestep during both training and prediction. Because of this, temporal information is only used implicitly instead of being modeled as an evolving sequence over time. This is also the method that is being applied in the paper by Cué La Rosa et al., 2025. This thesis will try to expand on this method.

Using an explicit spatio-temporal model for predicting near-future deforestation has not been fully explored yet. The model evaluation in Cué La Rosa et al., 2025 has shown that aggregating the temporal information performs well in areas where deforestation is already taking place, but it tends to underperform in areas where there is no recent history of deforestation. Additionally, while recent projects such as ForestCast show how important the historical context is for accurate predicting (Overlan et al., 2025), the optimal way to explicitly model these temporal dynamics is still an active area of research. This highlights the need for models that explicitly learn the temporal patterns and dynamics, instead of relying only on the aggregated historical data. The goal of this thesis is to build further on the Forest Foresight project by explicitly modeling the temporal evolution of deforestation using deep learning models. By investigating different model architectures and temporal windows, this thesis hopes to improve the predictive performance and, as a consequence support more effective forest protection.

2 Related work & Research Needs

2.1 Related Work

The importance of tropical forest monitoring and anticipating new locations of deforestation is highlighted above. However, there is still progress to be made in the predictive modeling of deforestation. The current state of the art can be divided into three main approaches: rule-based models, classical machine learning models, and deep learning models (Dominguez et al., 2022; Mayfield et al., 2020; Saha et al., 2020). Rule-based models are often used as a baseline or within more complex workflows. They are simple to interpret but tend to oversimplify real-world processes, which limits their effectiveness in predictive applications (Rosa et al., 2013). Examples include logistic regression, generalized linear models, and decision analysis techniques (Bera et al., 2022; Mets et al., 2017).

Classical machine learning approaches, such as Random Forest (RF), Support Vector Machine (SVM), and Multilayer Perceptron (MLP), have demonstrated strong performance in handling high-dimensional and non-linear data (Boateng et al., 2020). However, these models often struggle to effectively capture spatial and temporal dependencies (Hengl et al., 2018; Ouma et al., 2023).

More recently, deep learning models such as Convolutional Neural Network (CNN)-based architectures have been applied to deforestation detection and prediction using remote sensing data (de Andrade et al., 2022; De Bem et al., 2020; Kanoga et al., 2024; Lee et al., 2020; Md Jelas et al., 2024; Wang et al., 2023). These models can process complex, large-scale spatial and temporal information and can achieve high predictive accuracy. However, their performance strongly depends on model architecture and context. They typically require substantial training data and computational resources, and their interpretability is often limited (Kumar et al., 2023; Luo & Xu, 2023).

Deep learning models are capable of capturing temporal patterns and dependencies. However, many deforestation prediction studies rely on static or temporally aggregated input features rather than explicitly modeling temporal sequences. Although recent approaches such as the Forest Foresight model (Cué La Rosa et al., 2025) incorporate temporal information through aggregated features (e.g., accumulated forest disturbance over the past months), they operate on a single current timestep during both training and prediction. Because of this, temporal information is only used implicitly instead of being modeled as an evolving sequence over time. Model evaluation has shown that this approach performs well in areas where deforestation is already taking place, but it tends to underperform in emerging deforestation frontiers where temporal dynamics are more important. This highlights the need for models that explicitly learn temporal patterns, instead of relying only on aggregated historical context.

2.2 Research Need

Because of the limitations described above, many existing models focus mainly on the spatial dimension of deforestation while the temporal dynamics are underrepresented (De Bem et al., 2020; Masolele et al., 2021). Deforestation is driven by both environmental and human factors, and its progression over time is dynamic. Including explicit temporal information and patterns can lead to more realistic predictions of how deforestation progresses.

Spatio-temporal deep learning architectures such as ResUNet variants, CNN-LSTMs, 3D-CNNs, and attention-based models have shown improvements over purely spatial models (B.

Liu et al., 2025; Maretti et al., 2021; Md Jelas et al., 2024). The goal of this thesis is to build on this development by explicitly modeling the temporal evolution of deforestation using deep learning.

3 Research Questions

The Forest Foresight model in Cué La Rosa et al. (2025) was developed to predict near-future tropical deforestation. This thesis aims to further investigate the temporal aspect of this deforestation prediction. This will be done by investigating how different temporal input windows and prediction horizons influence model performance. In doing so, this thesis hopes to improve predictive performance and thereby support more effective forest protection and management.

Main Research Question:

What is the added value of explicitly modeling temporal dependencies for improving the accuracy and generalization of spatio-temporal deep learning models for tropical deforestation prediction?

Sub-questions:

- Which deep learning architecture is most suitable for modeling spatio-temporal patterns of deforestation?
- What is the optimal temporal window size for capturing relevant deforestation dynamics in predictive models?
- How do different prediction horizon's (for example 1, 3 or 6 months) influence model performance?
- How well does an improved temporal model for Gabon generalize to other countries, for example Venezuela and Papua New Guinea?

4 Data & Methods

This section describes the methodology that was applied for this research to predict near-future tropical deforestation. Figure 1 shows an overview of the pipeline. It is divided into four phases: data collection, preprocessing, modeling and evaluation. These steps will be discussed in the sections below.

4.1 Study Area

For this thesis, Gabon is chosen as the study area (see Figure 2). The reason for this is that the current Forest Foresight model is already operational in Gabon. It has a strong baseline performance, there is sufficient dense tropical forest in Gabon and it has suitable data availability. The analysis for the deforestation prediction will be restricted to forested areas only, for example as is defined by Turubanova et al. (2018). As secondary regions, Venezuela and Papua New Guinea are selected to evaluate the model generalization. This was chosen to test the model in areas on different continents than Gabon.

4.2 Data

4.2.1 Dynamic Variables

The dynamic variables are updated every month. There are multiple aggregated monthly variables available, such as the sum of deforestation in the last 6 months, last 3 months or last month. In the context of this thesis, only the last month variable is used for the 3D models. For the 2D model, the aggregated monthly data is used. The source of this information is the GFW alerts, which combines GLAD, GLAD-S2 and RADD alerts (Reiche et al., 2024). Next to that, precipitation and temperature are dynamic variables that are used for this thesis. Both the variables contain predictions of the climate for the coming 6 months. This information is sourced from CMIP6, and it is the median of 16 climate scenario's in which precipitation is expressed in mm/month and temperature in Kelvin (Eyring et al., 2016). Additionally, the historical context of deforestation is also used as dynamic data. This is done in the variables `timesince loss` and `total loss alerts`. `Timesince loss` is the time since the most recent deforestation event, and the `total loss alerts` is the sum of all deforestation events in that pixel since the start of the dataset.

4.2.2 Yearly Variables

Next to dynamic variables that are updated every month, there are also some variables that are updated with a longer interval. These variables are classified as yearly variables. There are two notable yearly variables that are being used for this thesis. These are the `closenesstoroads` variable and the `losslastyear` variable. `Closenesstoroads` is a variable that describes the distance to the nearest road per pixel. The source for this information is Open Street Map ("OpenStreetMap", n.d.) vector data. Only roads that are classified as 'Way' or 'Highway' are considered for this feature. To calculate the distance, road lines are simplified to 10 meters. The distance per pixel is calculated and transformed with a logarithmic formula:

$$pixelvalue = 255 - 20 * \log(distance + 1) \quad (1)$$

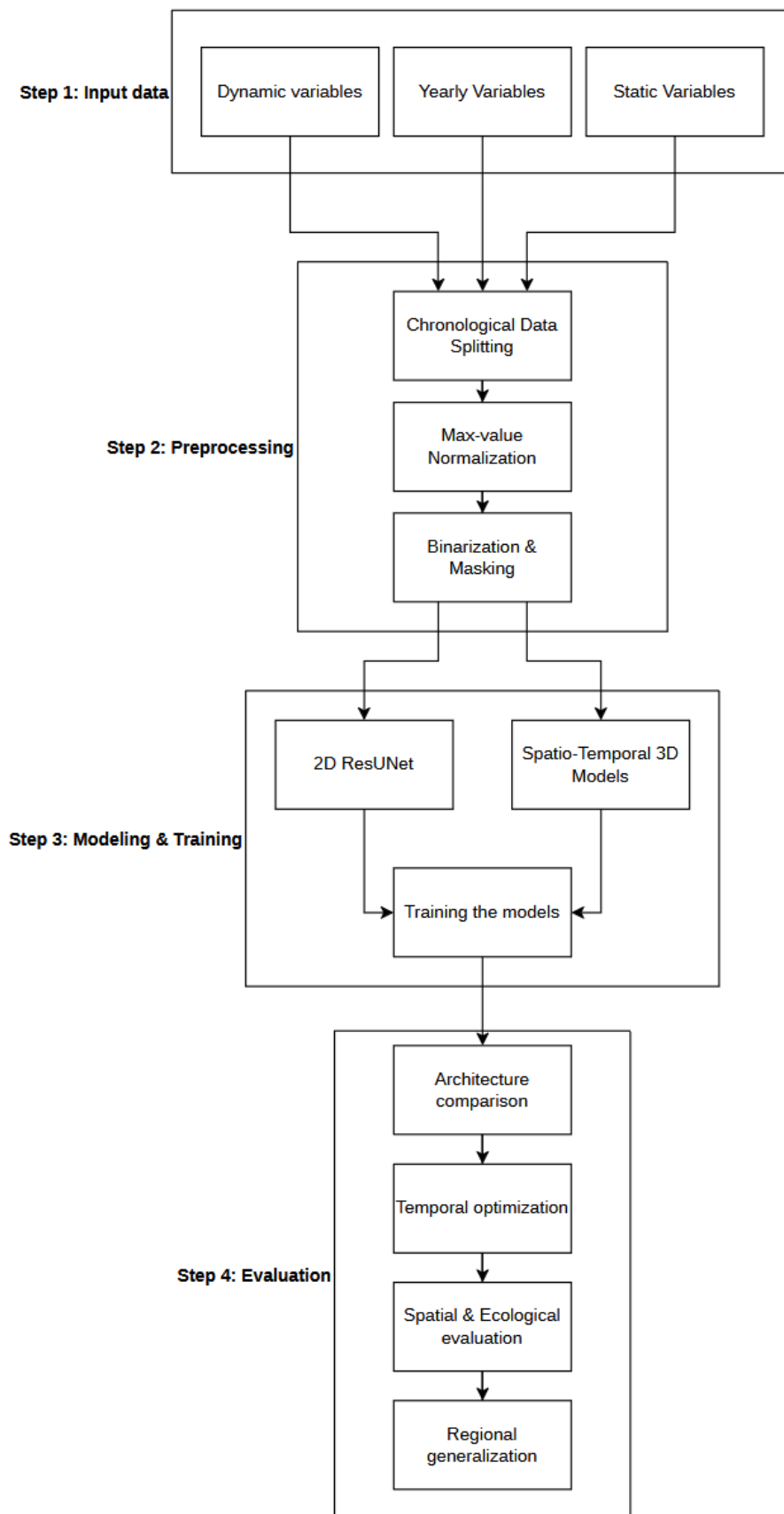


Figure 1: Methodological flowchart illustrating the four main steps of the research pipeline.



Figure 2: The location of Gabon in Africa

Losslastyear is based on the GLAD alerts. This is a dataset that is originally in a resolution of 30x30m. The losslastyear is the sum of all 30x30m pixels that were deforested last year in the 400x400m pixel.

4.2.3 Static Variables

The static variables mainly consist of ecological, geographic or demographic features. These are features that do not tend to change much over time, so a single snapshot was used for them to capture their information. One of the geographic features that is used is the topography. This is expressed in the features elevation and slope. Elevation is expressed in meters above sealevel, and slope as the angle of the slope (Farr et al., 2007). Next to this, the features xy and landpercentage also describe the geography. The xy features describes the lat/lon coordinates, and the landpercentage features describes the fraction of the pixel that is not water.

For ecological variables, there are the wetlands, peatland, catexcap and forestheight variables. peatland is the average depth of the peatland, with a maximum value of 15 meters. catexcap stands for the Cation Exchange Capacity. This is an indicator for soil health, which is useful as a proxy to determine the potential of agriculture. The wetlands variable is a binary variable that describes if there is any wetland within the pixel. Lastly there is the forest height variable. This gives the average height of trees within the pixel in 2020. Other ecological variables that are used consist of historical forest cover. These variables are historicloss and initialforestcover. Historic loss is defined as the forest that has been lost between 2001 and 2018. The initialforestcover is the forest cover that was present in 2019. Additionally, some variables that consider the agricultural potential are also used. These variables are calculated using artificial Neural Networks that are based on the topography, climate and accessibility of the area. These variables include croplandcapacity100p, below50p and over50p (Čengić et al., 2023).

Finally, some infrastructure and demographic variables are also taken into account. These include closenesstowaterways, populationcurrent, populationincrease and wdpa. All the used variables and their maximum values can be seen in Table 1.

Table 1: Overview of all dynamic, yearly, and static variables used in the 3D models, including their maximum values used for max-value scaling.

Category	Variable Name	Max Value
Dynamic	lastmonth	1600
Dynamic	precipitation	240
Dynamic	temperature	3000
Dynamic	timesinceloss	10000
Dynamic	totallossalerts	1600
Dynamic	month	12
Dynamic	sinmonth	1
Yearly	closenesstoroads	255
Yearly	losslastyear	256
Static	closenesstowaterways	255
Static	elevation	8849
Static	historicloss	256

Continued on next page

Table 1 continued from previous page

Category	Variable Name	Max Value
Static	initialforestcover	10000
Static	populationcurrent	20000000
Static	populationincrease	20000000
Static	slope	4000
Static	wetlands	1
Static	peatland	15
Static	croplandcapacity100p	255
Static	croplandcapacitybelow50p	255
Static	croplandcapacityover50p	255
Static	landpercentage	254
Static	forestheight	50
Static	wdpa	1
Static	catexcap	1000
Static	longitude (from xy)	180
Static	latitude (from xy)	90

4.3 Preprocessing

Before the raw data can be used as information to train the models, it is necessary to first transform the raw images into a structured format that is suitable for machine learning models. The goal of the preprocessing step is to ensure data quality and create consistent training, validation, and testing datasets. To create these datasets, the dynamic, yearly, and static variables are used as input. The output of the preprocessing pipeline is a structured set of normalized features and their corresponding ground truth labels, which ensures the models can train efficiently and stably without being hindered by inconsistent data scales or unhandled missing values.

Because temporal information is being used, it is not possible to split the data randomly. Doing this would lead to data leakage, where information from the future is used to predict scenarios in the past. To prevent this from happening, it is important to maintain strict temporal splits. The filenames of the features contain information on the date on which it was generated. This information is used to divide the data chronologically into a training, validation, and test set. For the initial comparison of the model architectures, the training set consists of data from the years 2021 and 2022. The validation set spans from June 2023 to May 2024, and the independent test set covers the subsequent period from June 2024 to May 2025. This chronological separation ensures that the models are evaluated purely on their ability to forecast unseen future events based on historical patterns. The timeline of the splits can be seen in Figure 3.

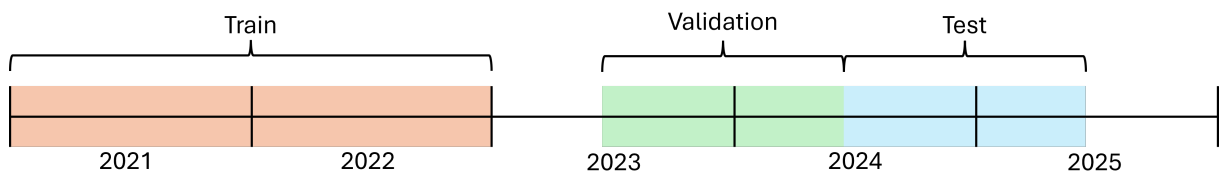


Figure 3: Timeline showing the train, validation and test splits

In the case that a specific variable is not available for the requested time period, the most recent file of that variable is used instead. So for example, if the precipitation of July 2024 is not available, but of June it is, then the precipitation of June will be used. Next to handling missing data, normalization of the data is applied to make sure that the deep learning models train in a stable manner. In this case, max-value scaling is applied. The max value is sourced from the metadata of the Forest Foresight dataset. Every pixel value across all data splits is then divided by this maximum value to scale the data to a range between 0 and 1. The function for this is described below:

$$X_{normalised} = X_{raw} / X_{max.train} \quad (2)$$

In case of outliers, so where the value after normalisation is greater than 1, the pixel value is clipped to 1. The target variables also have to be preprocessed. The standard ground truth files contain a value between 0 and 1600 per pixel. Every pixel in the ground truth maps has a resolution of 400x400m, but the ground truth value is based on 10x10m resolution images. This value between 0 and 1600 signifies the amount of underlying sub-pixels within that pixel that will be deforested in the coming x months. Because the prediction task for this thesis is a classification task rather than a regression task, it is necessary to convert the ground truth to binary values. For this reason, it was decided to classify every pixel with a ground truth value greater than 0 as deforestation (1), and pixels with a ground truth value of 0 as non-deforestation (0). Next to that, pixels that are outside of the region of interest, such as water bodies or pixels that are located in a different country, are masked out and they are not used during model training and inference.

4.4 Model Metrics

To determine the effectiveness of the trained models, it is important to measure it using metrics that explain how well the models perform for this specific task. In deforestation prediction tasks, the amount of deforested pixels is usually significantly lower than the amount of pixels that do not get deforested. In other words, there is a high class-imbalance. In cases like this, accuracy is a poor metric. If 95% of the pixels are not deforested, a model that always predicts no deforestation will achieve an accuracy of 95% without learning any features. It is more useful to look at the confusion matrix. This matrix describes true positives (TP), false positives (FP), true negatives (TN) and false negatives (FN). The confusion matrix can be seen in Figure 4. Based on these metrics, the precision, recall, precision-recall area under curve and $F_{0.5}$ score can be calculated.

Precision is the proportion of pixels that were predicted to be deforested, that were actually deforested. In the context of deforestation prediction, a low precision means that the model has many false alarms. A large portion of the pixels that are predicted to be deforested, do in reality not get deforested. Precision can be used as a measure for the reliability of the predictions that the model makes. Precision is calculated using the following formula:

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

The counterpart of precision is recall. Recall is the proportion of actual deforestation events that are detected by the model. A model that has a low recall score misses a substantial part of the actual deforestation events. Recall is calculated using the following formula:

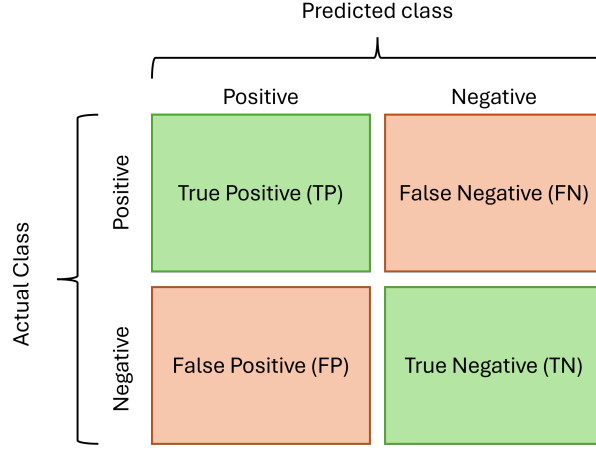


Figure 4: Confusion matrix

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

The main metric that is used to evaluate the model performance is the $F_{0.5}$ score. This is a metric that is based on the F-beta score. The F-beta score is a general metric that calculate the harmonic average of precision and recall, in which the user can determine which of the two should have more weights. The formula for this is as follows:

$$F_{\beta} = (1 + \beta^2) \cdot \frac{Precision \cdot Recall}{(\beta^2 \cdot Precision) + Recall} \quad (5)$$

By adjusting the β parameter, the balance between precision and recall can be adjusted. This function is most often used with $\beta = 1$, when it is called the F1-score. In this scenario, precision and recall have the same importance. When $\beta > 1$, the recall weighs more heavily on the score. A high recall will give a higher F_{β} -score than a similarly high precision. This goes the other way as well, when $\beta < 1$. In this case precision will weigh more heavily on the score. In this thesis, $\beta = 0.5$ is used. This means that in this $F_{0.5}$ score, the precision has a higher effect on the score than the recall. The reason for this is that in the specific use case of deforestation prediction, precision is more important than recall. A model with high precision but low recall generates a significant amount of false alarms. In practice, when the model is for example used by governments with limited supplies, this can lead to waste of time and resources. Models with a higher precision generate predictions that are more likely to be true, and the $F_{0.5}$ score reflects this practical need of a high precision.

$$F_{0.5} = (1 + 0.5^2) \cdot \frac{Precision \cdot Recall}{(0.5^2 \cdot Precision) + Recall} \quad (6)$$

Lastly, the models all use a dynamic threshold. Each model generates a value between 0 and 1 for every pixel that it predicts. The closer the predicted value is to 1, the more likely it is to be deforested. However, since predicting deforestation is a binary task, it is important to set a threshold above which a pixel is classified as 'will be deforested'. In this project, this threshold is based on the highest $F_{0.5}$ score that was reached in the validation set.

4.5 Models

In this thesis, multiple deep learning models are evaluated on their capabilities to use temporal data for near-future deforestation prediction. The 2D ResUNet from Cué La Rosa et al., 2025 is used as a temporally-aggregated baseline. To test the added value of explicit temporal modeling, this baseline model is compared to three spatio-temporal models: a 3D ResUNet, a 3D ConvLSTM, and a Video Vision Transformer (ViViT). When comparing the baseline with the 3D models, a methodological choice was made. Instead of doing a study where only the dimensions of the input data are changed, this research compares optimized models. The 2D ResUNet follows the architecture of the current Forest Foresight project such as specific design choices like using bilinear interpolation for upsampling. On the other hand, the 3D architectures use learnable transposed convolutions, which is the standard method for explicitly learning spatio-temporal patterns during upsampling. This approach makes sure that both the 2D baseline and the 3D models are evaluated at their maximum capability, which allows for a realistic comparison of their real-world predictive performance. The models differ in their complexity, and the number of trainable parameters per model can be seen in Table 2.

Table 2: Comparison of model complexity based on the number of trainable parameters.

Model Architecture	Trainable Parameters
2D ResUNet	3,032,993
3D ResUNet	9,084,225
ConvLSTM3D	9,102,497
ViViT	5,066,977

4.5.1 2D ResUNet

The 2D ResUNet (Diakogiannis et al., 2020; He et al., 2015; Ronneberger et al., 2015) is used as the temporally-aggregated baseline for this research. The model follows an encoder-decoder structure with residual connections. Unlike the 3D models which process separate files for each month of historical deforestation, this baseline model uses a single variable. This variable aggregates all the deforestation events of the past months into a single spatial image. Because of this, the input for this model is entirely 2D.

The data pipeline uses an 'early fusion' step to reshape the data, which keeps it compatible with the preprocessing pipeline that generates files in a 3D format. However, the model itself remains a standard 2D CNN that processes a tensor of shape $[C, H, W]$. The C in this tensor stands for the variables or 'channels' that are used as input, and the H and W stand for the height and width of the images. This means the network has spatial context, but it views the historical data as a static representation instead of an explicit temporal sequence (Cué La Rosa et al., 2025). To stay in line with the Forest Foresight architecture, bilinear interpolation is used for upsampling (Cué La Rosa et al., 2025) and a spatial dropout layer ($p = 0.3$) is added before the final classification to prevent overfitting in the training phase.

4.5.2 3D ResUNet

To explicitly model the temporal aspect of deforestation, the 3D ResUNet builds on the 2D architecture by using 3D convolutions (Çiçek et al., 2016). Instead of the 2D model, the 3D ResUNet processes the full spatio-temporal tensors with a shape of $[C, T, H, W]$. Here the T

stands for the time dimension. The model uses cube-shaped filters that move over the height, width, and time dimensions to learn spatio-temporal patterns.

An important detail of this architecture is that the encoder only applies downsampling on the spatial dimensions (H, W). This makes sure that the temporal dimension (T) is preserved throughout the deeper layers of the network. Compared to the 2D baseline, the decoder uses 3D transposed convolutions to restore the spatial resolution. These transposed convolutions are a bit more complex than the bilinear interpolation, and it allows the model to 'learn' the best method of restoring the images. The time dimension is maintained intact until the final classification layer. There it is collapsed by applying a 3D convolution with a temporal kernel size that is equal to the entire time depth. The effect of this is that the sequence is reduced into a single 2D prediction map. By doing this, the explicit temporal context is maintained until the final prediction step.

4.5.3 3D ConvLSTM

Conceptually, the 3D ConvLSTM is a hybrid model that combines the spatial pattern recognition of CNNs with the sequential memory of Recurrent Neural Networks (RNNs) (Shi et al., 2015). In this thesis, the architecture consists of a 3D encoder, a ConvLSTM bottleneck, and a 2D decoder. As the network steps through the months, the ConvLSTM cells use a hidden state to remember temporal dependencies, while still preserving the spatial structure of the features. A specific detail of this implementation is the use of asymmetrical skip-connections. These connections send spatial features exclusively from the last time step of the encoder directly to the decoder. This gives extra priority to the most recent state of the forest, which is useful for near-future forecasting.

4.5.4 ViViT

The Video Vision Transformer (ViViT) differs the most from the other models. Unlike the CNN-based models, the ViViT relies on self-attention mechanisms to model the spatio-temporal relationships (Arnab et al., 2021). To process the input data, the ViViT divides the 3D data into many smaller, non-overlapping blocks called 'tubelets'. After that, attention scores are calculated between all the tubelets. By doing this, the model can capture global dependencies across the entire time window, without having a localized view like CNNs.

To save computation time, this attention calculation is split into two parts. First, a spatial transformer processes the tokens within each individual frame. After that, a temporal transformer models the relationships across the frames. Finally, a series of transposed convolutions scales the processed data back to a prediction map.

4.6 Processing

The dataset that is used for deforestation detection is characterized by extreme class-imbalance. The majority of the pixels in the dataset are stable forest, and in general, deforestation events are rare. Without adjustments, the models would probably converge to a solution where they always predict the 'no deforestation' class, since this is the most occurring scenario.

To tackle this class imbalance for the deep learning models, a combined loss function was used. The loss function consists of a Binary Cross-Entropy (BCE) Loss and a Dice Loss (Milletari et al., 2016).

The BCE component evaluates the classification probability per pixel and the Dice Loss optimizes the spatial overlap between the predicted deforestation and the ground truth. Because Dice Loss evaluates the intersection over union of the positive predictions, it penalizes the model if it misses the rare deforestation events. This combination forces the network to identify the minority pixels.

Because of the high required computation capacity of the 3D spatio-temporal convolutions, the models are being trained on GPU's. The models are optimized using a Rectified Adam (RAdam) optimizer (L. Liu et al., 2021). The RAdam optimizer is chosen because it causes a more robust and stable convergence during the early stages of the training process. Additionally, other hyperparameters are set to specific values that are available to use on the available GPU-memory. These include for example a specific learning rate of 0.0001 and a batch size of 8.

More complex architectures like the 3D ResUNet have a very high capacity, which comes with a risk of overfitting when the models are trained on a limited set of spatial data. To prevent the models from memorizing noise in the training data, early stopping is applied. After every epoch, the $F_{0.5}$ score is calculated for the independent validation set. When this validation score does not improve after a pre-set number of epochs, the training process is automatically ended and the weights of the best performing model (i.e. with the highest $F_{0.5}$ score on the validation set) are saved.

As describe before, the models produce a continuous value between 0 and 1 that signifies the probability of that pixel to be deforested within the coming x months via a Sigmoid-activation function. To translate these probability values to a binary classification, forested or deforested, a dynamic threshold for the probability value is determined. Every pixel with a value above the threshold is classified as deforested, and pixels below are predicted to remain forested. The value for this threshold is determined specifically for each model based on the max $F_{0.5}$ score on the validation set. This threshold value is then saved and applied for evaluation on the test set. By determining the threshold on the validation set, data leakage is prevented and the final test results are a representation of the model performance in the real world.

4.7 Analysis

To systematically answer the research questions, the evaluation of the different models is structured using a funnel approach. This strategy starts with a broad comparison of the architectures, after which the best performing model is selected and tested further with additional spatial, temporal and geographic analyses.

4.7.1 Comparison of Model Architectures

The first phase is focused on quantifying the added value of applying spatio-temporal deep learning models for deforestation predictions. The baseline 2D ResUNet is compared to the more complex 3D models that were introduced in Section 4.5. The performance of the models is evaluated using the metrics defined in Section 4.4, and as explained before, the $F_{0.5}$ score is used as the main metric.

To make sure that the reported scores are a fair representation of the model performance, each model is trained over multiple iterations. A number of random seeds is used to make sure that every iteration of every model sees the same data. Because of this paired setup,

differences in scores can be explained by the model architecture itself, instead of stochastic variations in the training data.

By using the 5 scores per model, a Friedman test can be used to test whether there is a significant difference in model performance. The Friedman test is a non-parametric statistical test designed to check for differences in scores between multiple models. For this test, it is not necessary that the data is normally distributed, which makes it well-suited for a small number of samples (like 5 runs) and performance metrics bounded between 0 and 1.

The outcome of the Friedman test is a p-value. If this p-value is lower than 0.05, it means that there is a statistically significant difference in performance between the models. To further investigate this difference and see exactly which models differ from each other, a Nemenyi post-hoc test is conducted. This test compares all models pairwise by calculating their average ranks across the different iterations. It determines a critical distance and if the difference in average rank between two models is larger than this distance, their performance is considered significantly different.

Relying only on the p-value of the Nemenyi test can be limiting, especially in situations with small sample sizes (e.g., $N = 5$). Because the $F_{0.5}$ metric is bounded between 0 and 1, parametric testing is not suitable. However, with 5 paired observations, the minimum possible p-value for a non-parametric test is mathematically bounded at 0.0625. This makes it impossible to achieve a $p < 0.05$ significance level for models that rank next to each other, regardless of how large the difference in scores is.

Because of that, Hedges' g for paired samples is applied as an additional metric. This metric calculates the effect size of differences between samples. It is an adaptation of Cohen's d and applies a correction factor to prevent the overestimation of effect sizes in small samples. An effect size of $g \geq 0.8$ is considered a large difference. By combining the Nemenyi test with Hedges' g , the evaluation of the $F_{0.5}$ scores can be measured more effectively.

4.7.2 Temporal Context & Horizon

After the initial comparison of the different models, the best-scoring model is selected for temporal optimization. To determine how much historical information is required for the most accurate predictions, the model is trained using different context lengths of 3, 6, and 12 months. Additionally, different prediction horizons are evaluated to investigate how predicting over longer periods of time impacts model performance. The models are trained to predict over 3, 6, and 12 months into the future.

Evaluating a 12-month prediction horizon requires ground truth data exactly 12 months after the prediction date. For example, if a prediction has to be made for January 1st 2025, the ground truth needs to be available for January 1st 2026. In the original dataset splits describe above, the test period extended until June 2025. Predicting 12 months into the future for the final months of the test set would require ground truth data up to June 2026. Because this research was conducted in March 2026, this future data was not yet available, making the original test split unusable for this specific horizon analysis.

To accommodate this limitation, all temporal splits (training, validation, and testing) were shifted backwards by exactly three months compared to the original setup for the 12 month horizon analysis. This shift ensures that the necessary 12-month future ground truth data is available for the entire test set until March 2026. The effect of this adjustment is a training

set that is three months shorter. However, the validation and test sets still cover the same 12-month duration as in the other experiments. Specifically, the new validation targets span from April 2023 to March 2024, and the independent test targets span from April 2024 to March 2025.

4.7.3 Ecological & Spatial Evaluation

To translate the relatively abstract machine learning metrics to practical and real world situations, the most effective model is applied to spatial and ecological analyses on the test set.

- **Seasonal Analysis:** Deforestation patterns are influenced by the raining and dry seasons (Staal et al., 2020) and to get a better insight into this seasonality, the model is evaluated per calendar month. By looking at the scores per month, the temporal stability of the model can be quantified.
- **Spatial Distance Analysis:** An important requirement for deforestation warning systems is the ability of the model to predict new deforestation frontiers, instead of just predicting deforestation within the bounds of existing frontiers. A spatial analysis is made based on the deforestation history of each pixel. The $F_{0.5}$ score, precision and recall will be calculated for pixels that did not experience any deforestation in the preceding 6 months.
- **Patch Size Analysis:** In the real world, deforestation varies from small slash and burn practices to larger industrialized deforestation. By grouping connected pixels in the ground truth dataset, the recall can be calculated per category (small patches: < 10 pixels, medium patches: $10 - 49$ pixels, large pixels > 49). This will show how consistent the model performs over multiple spatial scales.

4.7.4 Model performance in Different Regions

To evaluate the spatial generalization of the best performing architecture from the initial evaluation, the models will be trained and evaluated on two new regions: Venezuela and Papua New Guinea. Instead of using the weights from the model that was trained on Gabon, the models for these two regions will be trained from scratch using regional data. This allows for a targeted evaluation of the architecture's capability to learn regional deforestation dynamics. To ensure that the results are comparable with the model described by (Cué La Rosa et al., 2025), the models are trained with a 6-month historical context and a 6-month prediction horizon. Just like for Gabon, the model performance will be evaluated using the $F_{0.5}$ score as the primary metric, and the same train, validation, and test splits are used. The goal of this analysis is to determine whether similar results are achieved outside of the study area of Gabon.

5 Results

5.1 Comparison of Model Architectures

The initial phase of the research compared the baseline 2D ResUNet model against three spatio-temporal architectures: 3D ResUNet, ConvLSTM3D, and ViViT. All models used a historical context window of 6 months and a prediction horizon of 6 months. Table 3 presents the $F_{0.5}$ scores on the test set across five iterations with fixed random seeds.

The 2D ResUNet achieved the highest mean $F_{0.5}$ score (0.6526). The ResUNet3D and ConvLSTM3D demonstrated similar but slightly lower performances, with mean scores of 0.6466 and 0.6457, respectively. The ViViT model recorded the lowest mean score across all iterations (0.6177).

Table 3: $F_{0.5}$ scores on the test set for the evaluated model architectures across 5 iterations. Best mean performance is highlighted in bold.

Model	Seed 2	Seed 3	Seed 4	Seed 5	Seed 6	Mean
2D ResUNet	0.6494	0.6563	0.6571	0.6519	0.6484	0.6526
ResUNet3D	0.6508	0.6478	0.6469	0.6415	0.6458	0.6466
ConvLSTM3D	0.6446	0.6447	0.6457	0.6461	0.6475	0.6457
ViViT	0.6173	0.6127	0.6170	0.6198	0.6217	0.6177

A Friedman test indicated a statistically significant overall difference in performance between the evaluated architectures ($\chi^2 = 12.12$, $p = 0.007$). A following post-hoc Nemenyi test revealed that the ViViT model performed significantly worse than the 2D ResUNet ($p = 0.0034$), with a large effect size (Hedges' $g = 4.10$). No statistically significant differences were found between the three convolutional models (2D ResUNet, 3D ResUNet, and ConvLSTM3D), with all pairwise comparisons yielding $p > 0.05$. For all following analyses in this research, the ResUNet3D model was used, because it scored the highest $F_{0.5}$ score of the 3D architectures.

5.2 Temporal Context Length

Table 4 presents the $F_{0.5}$ scores for the ResUNet3D model when varying the temporal input sequence length (3, 6, and 12 months). The 3-month historical sequence achieved the highest mean $F_{0.5}$ score (0.6503), followed closely by the 12-month (0.6489) and 6-month (0.6466) sequences.

Table 4: $F_{0.5}$ scores on the test set for the ResUNet3D model utilizing different temporal context lengths.

Context Length	Seed 2	Seed 3	Seed 4	Seed 5	Seed 6	Mean
3 Months	0.6505	0.6485	0.6478	0.6470	0.6578	0.6503
6 Months	0.6508	0.6478	0.6469	0.6415	0.6458	0.6466
12 Months	0.6509	0.6492	0.6366	0.6559	0.6519	0.6489

A Friedman test yielded no statistically significant difference between the context lengths ($\chi^2 = 2.80$, $p = 0.2466$). Furthermore, the calculated Hedges' g effect sizes between these conditions were negligible.

5.3 Prediction Horizon

To investigate the impact of forecasting over longer periods, the ResUNet3D model was evaluated across prediction horizons of 3, 6, and 12 months, using a fixed 3-month historical context. As shown in Table 5, performance increased with longer horizons. The 12-month horizon achieved the highest mean $F_{0.5}$ score (0.7222), while the 3-month horizon recorded the lowest (0.5068).

Table 5: $F_{0.5}$ scores on the test set for the ResUNet3D model across different prediction horizons.

Prediction Horizon	Seed 2	Seed 3	Seed 4	Seed 5	Seed 6	Mean
3 Months	0.5366	0.4688	0.5133	0.4996	0.5159	0.5068
6 Months	0.6505	0.6485	0.6478	0.6470	0.6578	0.6503
12 Months	0.7234	0.7210	0.7211	0.7204	0.7253	0.7222

A Friedman test confirmed a statistically significant difference between the horizons ($\chi^2 = 10.00$, $p = 0.0067$). A post-hoc Nemenyi test indicated a significant difference between the 3-month and 12-month horizons ($p = 0.0045$). While adjacent intervals (e.g., 3-month vs. 6-month) did not show statistical significance ($p = 0.2538$), the effect sizes were large. The performance improvement from a 3-month to a 6-month horizon resulted in an effect size of $g = 4.80$, and the transition from 6-month to 12-month yielded $g = 23.05$.

5.4 Spatio-Temporal Performance and Regional Generalization

To gain detailed insights into specific model behaviors and to assess the spatial generalization of the temporal dimension, the 3D ResUNet and 2D ResUNet were evaluated on the test year using 6 months of context and a prediction horizon of 6 months across three different study areas: Gabon, Papua New Guinea (PNG), and Venezuela. Figure 7 shows the effect of seasonality, patch size, and the focus on emerging frontiers for both architectures across these three regions.

First, the models were evaluated globally across the 12 months of the test year to observe temporal variability (Figure 7, top row). In Gabon, both architectures showed similar seasonal patterns. In Papua New Guinea, the baseline 2D ResUNet consistently matched or slightly outperformed the 3D architecture throughout the year, peaking at an $F_{0.5}$ score of 0.706 in October. In Venezuela, overall performance was lower for both models ($F_{0.5}$ ranging between 0.43 and 0.53). A paired Wilcoxon signed-rank test across the 12 months confirmed no statistically significant difference in global performance for Gabon ($p = 0.110$) and PNG ($p = 0.791$). However, in Venezuela, the models did show a statistically significant difference ($p = 0.021$).

Furthermore, the sensitivity of the models to the spatial extent of deforestation was analyzed (Figure 7, middle row). Across all three regions, the average recall improved sequentially with larger patch sizes. In Gabon and PNG, both architectures demonstrated similar capabilities in capturing spatial patterns, with recall scores surpassing 0.65 for patches larger than 50 pixels. Interestingly, in Venezuela, the models were very sensitive to larger patches, occasionally reaching recall rates above 0.70. A Wilcoxon signed-rank test revealed no statistically significant differences ($p > 0.05$) between the 2D and 3D architectures across any of the patch size categories for Gabon and PNG, indicating no distinct spatial advantage for either approach.

in general detection. The only exception was observed in Venezuela for large patches (≥ 50 pixels), where the 2D model achieved a statistically significant improvement in recall over the 3D architecture ($p = 0.027$).

Finally, the models were evaluated exclusively on emerging frontiers to test their ability to predict completely new deforestation events (Figure 7, bottom row). These frontiers were defined as forested pixels that experienced no deforestation in the preceding 6 months. In Gabon, explicitly modeling the temporal dimension gave a significant advantage: the 3D ResUNet identified 42.04% of all new deforestation events, resulting in a notably higher $F_{0.5}$ score (0.601) compared to the 2D model (0.509). A paired Wilcoxon signed-rank test confirmed this improvement in overall $F_{0.5}$ score on emerging frontiers in Gabon to be statistically significant ($p < 0.01$).

However, when looking at the entire country, this advantage does not translate to a completely different predictive behavior. As shown in Figure 5, the general spatial distribution of the predictions for June 2025 remains comparable between the two architectures. Both models successfully identify the same large-scale deforestation patterns (green) and struggle with similar missed events (pink). This visual evidence supports the findings from the architecture comparison (Section 4.7.1), which showed that the 2D and 3D models achieve similar overall performance.

2D ResUNet predictions compared against 3D ResUNet predictions in Gabon

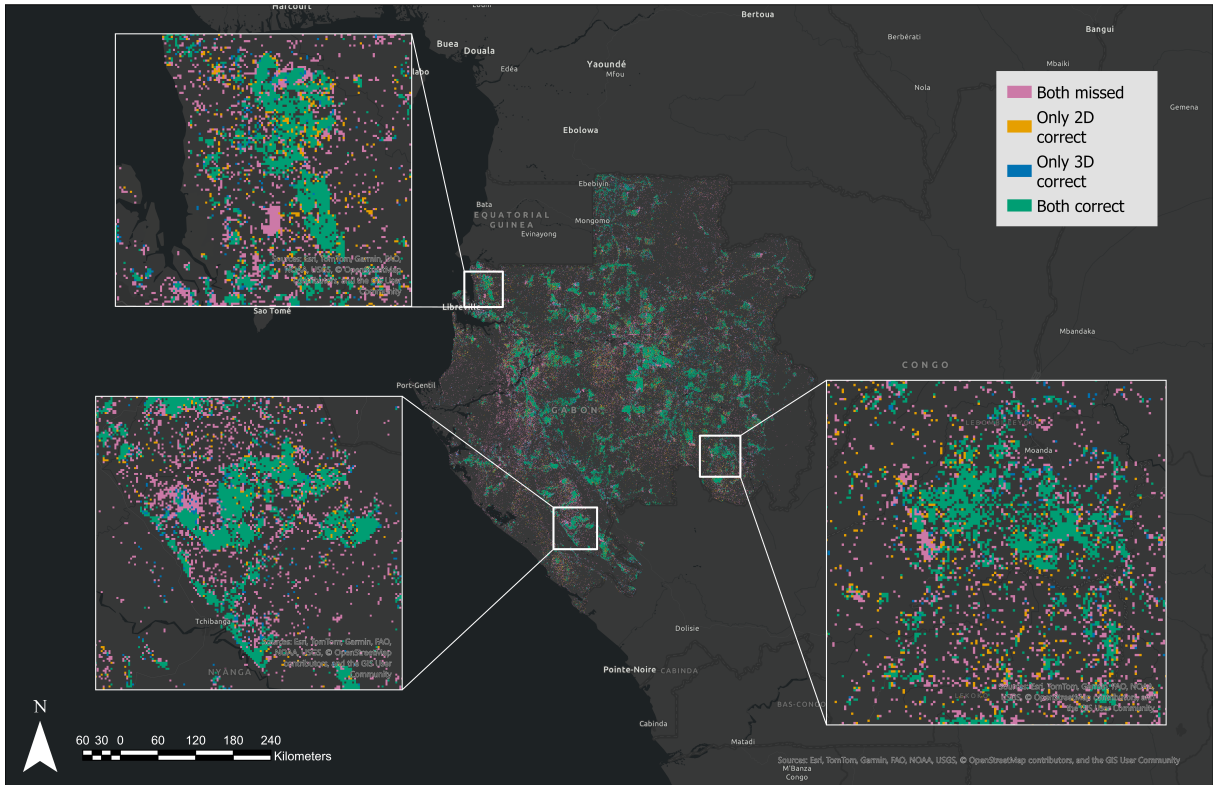


Figure 5: Spatial comparison of predicted deforestation frontiers in Gabon for June 2025 between the baseline 2D ResUNet and the spatio-temporal 3D ResUNet, using 6 months of context and a horizon of 6 months.

This new frontier dynamic was reversed in Papua New Guinea. The baseline 2D ResUNet significantly outperformed the 3D ResUNet, achieving a mean $F_{0.5}$ score of 0.571 compared to

0.483 for the 3D model. The 2D model achieved a strong average precision (0.671), whereas the 3D model suffered a substantial drop in recall (0.262), indicating that 3D convolutions introduced noise instead of predictive value in this specific study area. A Wilcoxon signed-rank test confirmed that the 2D model's better predictive performance on emerging frontiers in PNG was significant ($p < 0.01$).

These regional performance differences correspond to distinct spatial patterns in the ground truth and predictions. As shown in Figure 6, a visual comparison of the 3D ResUNet performance between Gabon and Papua New Guinea shows different deforestation dynamics. In Gabon, deforestation predominantly occurs in larger, continuous patches. In contrast, the distribution in Papua New Guinea is more fragmented.

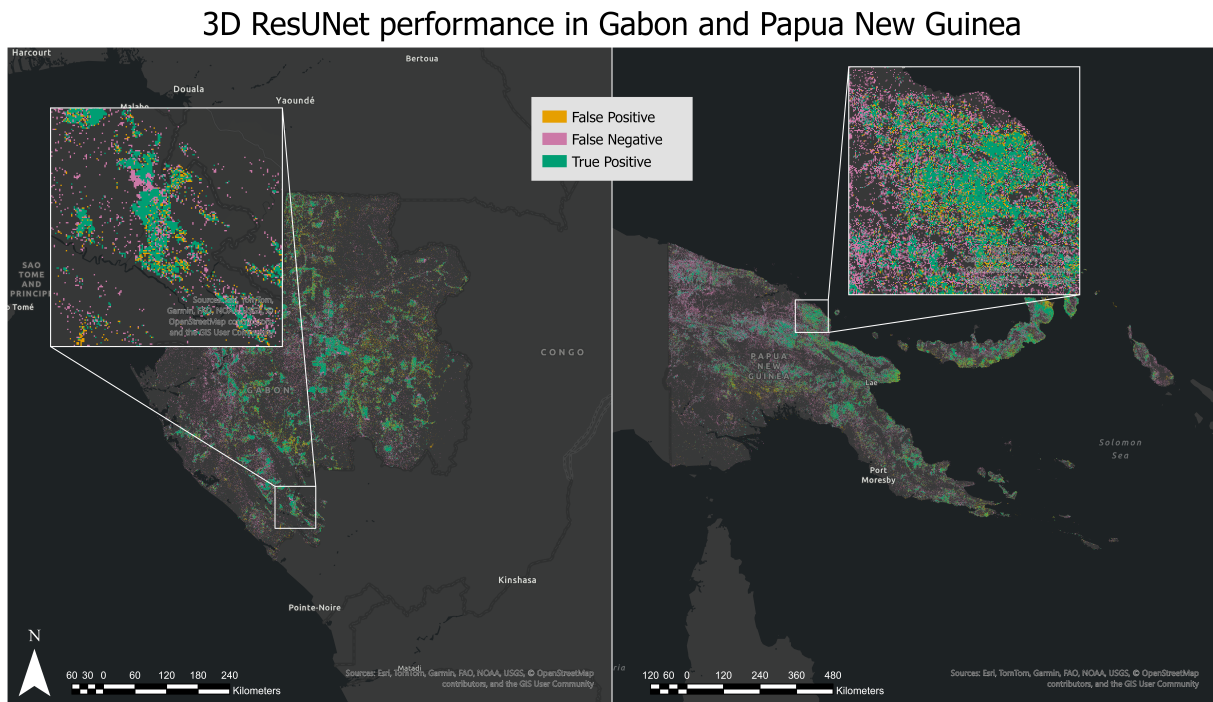


Figure 6: Spatial comparison of the 3D ResUNet performance in Gabon (left) and Papua New Guinea (right). The map shows the difference in deforestation dynamics in the two regions. Gabon has continuous patches of deforestation, while PNG shows more fragmented disturbances.

In Venezuela, the 3D ResUNet demonstrated a slight advantage over the 2D model ($F_{0.5}$ of 0.424 versus 0.395), primarily driven by a higher recall (0.443), though both models recorded low precision scores. Despite the lower absolute scores, a Wilcoxon test verified that the 3D model's advantage on Venezuelan emerging frontiers was also statistically significant ($p < 0.01$).

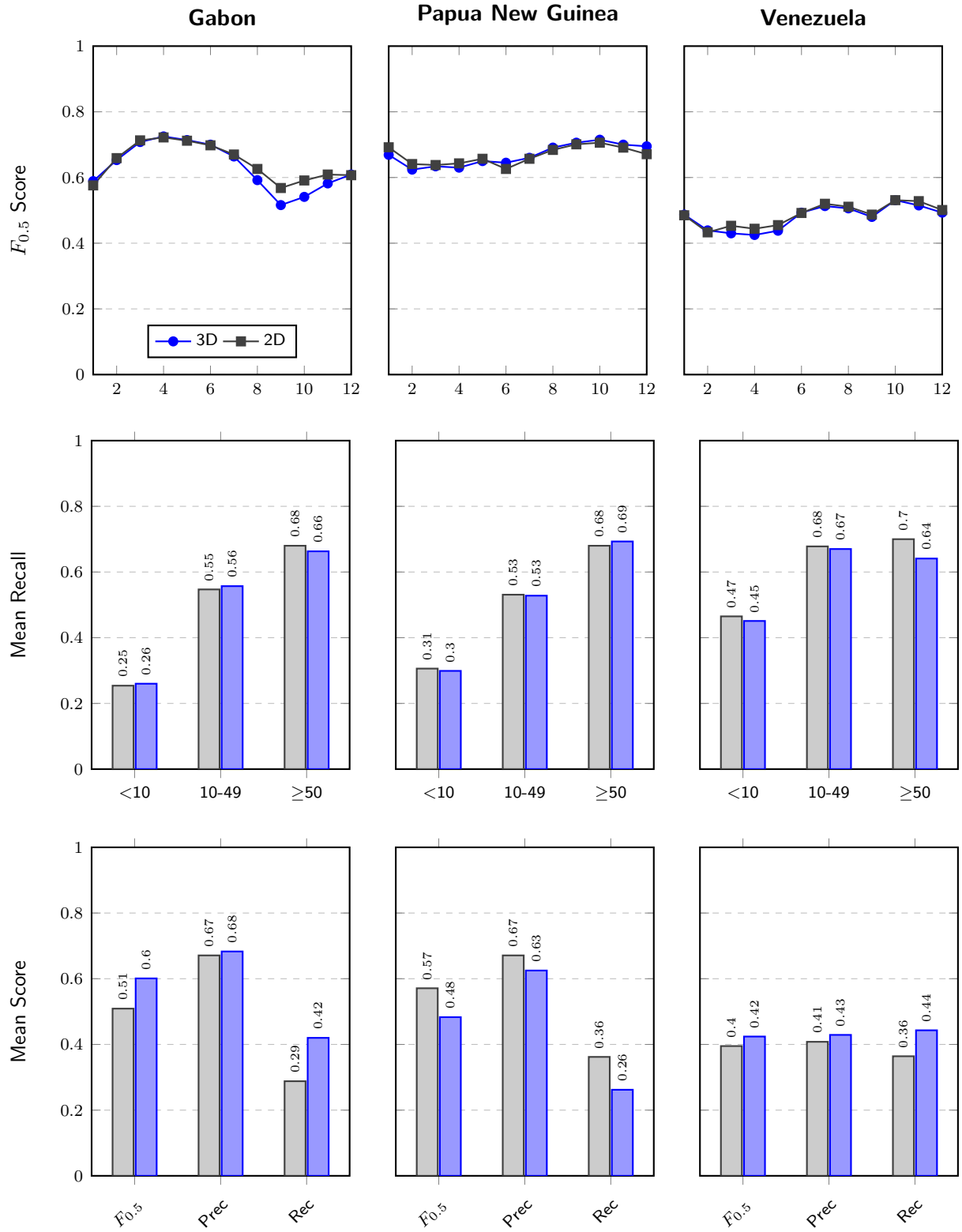


Figure 7: Performance comparison between the 2D (gray) and 3D (blue) architectures across Gabon, Papua New Guinea, and Venezuela. Top row: Global monthly seasonality showing $F_{0.5}$ scores. Middle row: Recall across different spatial patch sizes in pixels (12-month average). Bottom row: Predictive performance exclusively on new emerging frontiers (12-month average).

6 Discussion

6.1 The Added Value of Explicit Temporal Modeling

The primary objective of this thesis was to evaluate the added value of explicitly modeling temporal dependencies for tropical deforestation prediction. The most important finding of this research is that the overall results indicate that explicit spatio-temporal modeling does not provide a universal, significant improvement over temporally aggregated features. For the primary study area (Gabon) and Papua New Guinea, the overall performance across the 12-month test year showed no statistically significant difference between the two architectures. This suggests that for general deforestation forecasting, aggregating historical context into a single spatial feature map is sufficient, and the added computational complexity of 3D convolutions does not inherently result in better overall predictions. An explanation for the similar performance of the 2D and 3D models could be in the fact that the history of deforestation in the pixels is already captured sufficiently in the provided variables. Both models use features such as *losslastyear*, *timesince loss* and *totallossalerts*, which can give a clear picture of the past deforestation. It is hypothesized that because the historical context is already represented in these variables, explicitly analyzing the timeline of deforestation in 3D does not add significant extra predictive performance (Geirhos et al., 2020). However, without a feature importance analysis, this remains a hypothesis. Future research is needed to confirm the exact impact of these variables in 3D models.

In Venezuela, a statistically significant difference was observed globally in favor of the 3D model. However, the absolute performance in this region was low for both architectures, with the $F_{0.5}$ ranging between 0.43 and 0.53. Because of these overall poor results, the small statistical advantage of the 3D model might be caused by random variance or statistical noise instead of a structural architectural superiority. The shared difficulty to achieve accurate predictions in Venezuela can be explained by the specific deforestation dynamics that occurred during the evaluation period. The models in this research were trained on historical data from 2021 and 2022. However, according to Cué La Rosa et al. (2025), Venezuela experienced a sharp rise in forest loss in 2024 that was primarily driven by forest fires resulting from climate variability. Because the independent test set for this research covers the period from June 2024 to May 2025, it directly overlaps with these anomalous events. In Figure 8, the difference in the scale of deforestation between september 2022 and september 2024 can be seen. Deep learning models that rely heavily on historical spatial patterns and recent proximity to deforestation struggle to capture these sudden, seasonal fire events that deviate from the training data. The unpredictable nature of forest fires leads to a significant increase in false negative rates, which explains the poor overall predictive performance and low $F_{0.5}$ scores for both the 2D and 3D architectures in this region (Jayathilake et al., 2021; Shmuel & Heifetz, 2022; Staal et al., 2020).

When zooming in on the specific task of predicting entirely new deforestation events, the evaluation showed contrasting results. While the 3D model significantly outperformed the 2D model in Gabon on these new frontiers, this dynamic was completely reversed in PNG. It is important to note that emerging frontiers represent a small fraction of the total number of evaluated pixels. The contradicting results between the evaluated regions suggest that the effectiveness of 3D models on these specific events is very dependent on the local geographic situation, the local history of deforestation and regional deforestation dynamics. Additionally, because the models were evaluated on a single test year, it cannot be ruled out that the 3D model's superior performance on new frontiers in Gabon was coincidental or specific to the

Deforestation events in Venezuela during september 2022 and september 2024

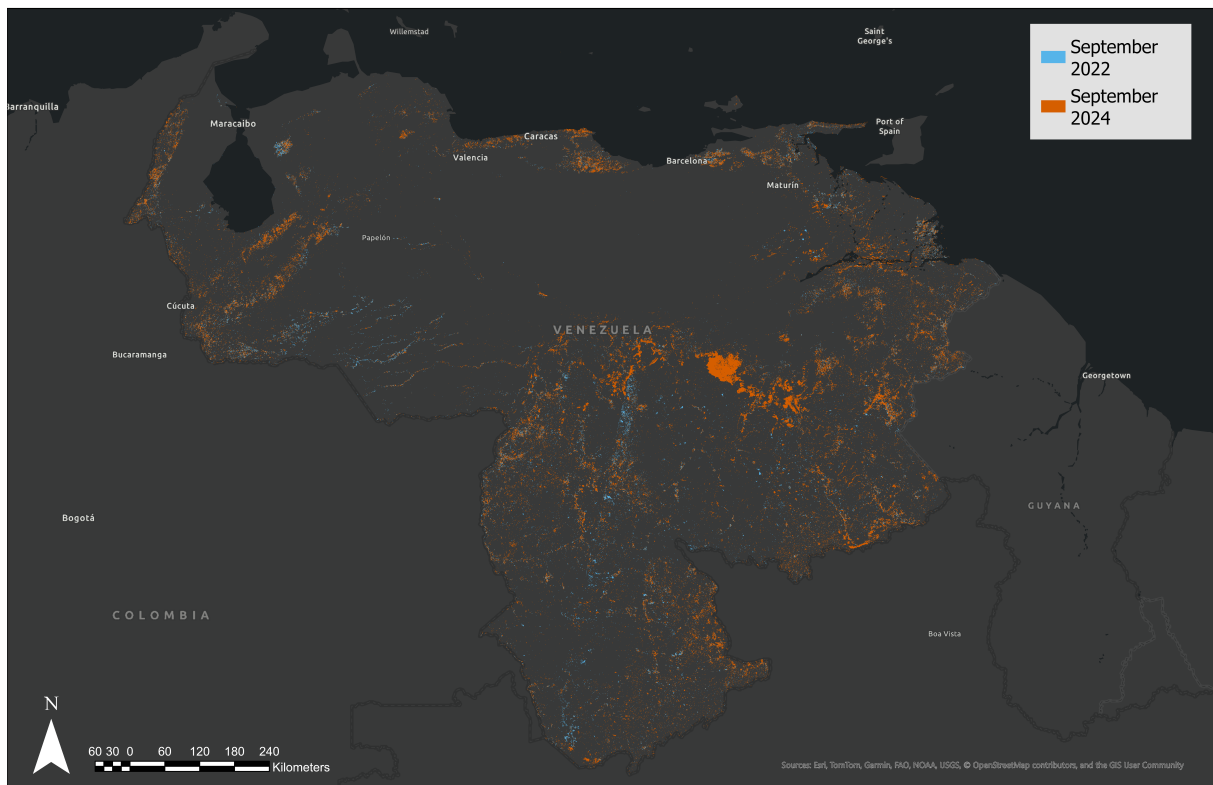


Figure 8: Spatial distribution of forest disturbances in Venezuela based on ground truth. The map shows the stable historical baseline of September 2022 (light blue) and the extreme scale of the September 2024 wildfires (orange). This shift in dynamics could explain the limited predictive performance of the models in this region.

temporal aspects of that particular year. In general, relying on explicit temporal sequences makes models more sensitive, which can either work as an advantage (e.g. Gabon) or introduce noise (e.g. PNG) for predicting new deforestation events.

6.2 Influence of Temporal Context and Prediction Horizon

Another finding relates to the influence of the prediction horizon on the model's performance. The results show that predicting over a 12-month horizon led to a significantly higher $F_{0.5}$ score of 0.7222 compared to the shorter horizon models of 6 and 3 months. Intuitively, it might seem more difficult to predict further into the future. However, the target variable for the models is the accumulation of deforestation events in the coming x months. Because of this fact, the ground truth of the 12-month horizon contains a larger amount of deforestation events than the 6- and 3-month ground truths. This provides an easier target for the models to predict. On the other hand, predicting the deforestation within the 3-month window requires a higher temporal precision, since there are less events within this time-frame, which makes it a significantly more difficult classification task. This explains the significant differences between the scores between the 3, 6 and 12 month horizon models.

When investigating the context length, the evaluation showed that the model with a shorter context length of 3 months performed slightly, but not significantly, better than the 6-month and 12-month context models. This implies that the most important drivers of near-future deforestation are already present in the short-term dynamic variables or the static or yearly variables. Deep learning models often struggle with longer sequences if older data introduces noise or redundant information (Albelali & Ahmed, 2025; Liang et al., 2025). The fact that the model reaches its highest $F_{0.5}$ score with only 3 months of history suggests that recent events are sufficient to predict future deforestation. This makes shorter context windows a more computationally efficient choice.

6.3 Comparison of Deep Learning Architectures

The comparison between the different model architectures revealed that all the models using convolutions (2D ResUNet, 3D ResUNet, ConvLSTM3D) performed similarly to each other and significantly outperformed the attention-based ViViT model. This difference in performance can be explained by the inherent differences in how these architectures process spatial relationships. Deforestation is a highly localized spatial process, it typically expands gradually from existing forest edges, newly built infrastructure, or nearby agricultural patches (Acheampong et al., 2019; Trigueiro et al., 2020). Convolutional Neural Networks have a high inductive bias, meaning that their sliding localized filters are explicitly designed to recognize nearby spatial patterns, textures, and borders by default (Diakogiannis et al., 2020; Li et al., 2022; Ronneberger et al., 2015). This makes them very suitable for detecting the spread of deforestation.

On the other hand, the ViViT architecture relies completely on self-attention mechanisms. This allows the model to capture a global overview and long-range dependencies by evaluating the relationships between all parts of the input simultaneously, but it does not have the localized inductive bias of CNNs. Because of this, the ViViT might struggle to distinguish the important local signals of a new deforestation edge from irrelevant and distant background information. Furthermore, transformer models require high volumes of data for training. Because they do not have built-in mathematical assumptions about spatial connections like CNNs do, they

require massive datasets to learn basic spatial structures (Arnab et al., 2021; Dosovitskiy et al., 2021; Kumar et al., 2023). Given the constrained temporal and geographic scope of this research and the extreme class imbalance where deforestation events are rare, the ViViT probably did not receive the necessary volume of diverse positive samples to effectively optimize its attention weights. This resulted in a lower predictive performance compared to the convolutional models.

6.4 Limitations of the Study

Despite the interesting results that were explained above, there are several limitations that should be acknowledged. First of all, the spatial resolution of the ground truth pixels in the dataset is fixed at 400x400 meters per pixel. For the ground truth, this 400x400m is composed of 1600 10x10m sub-pixels. The value of the 400x400m pixel is the sum of 10x10m pixels that experience deforestation in the coming x months. However, to process this information for the classification task in this thesis, this information was binarized. This means that all 400x400 m pixels that had a value greater than 0 were treated as a deforestation pixel. No differentiation was made between a pixel with only 1 event against a pixel with 1600 events. Both pixels will be classified as 1. This means that the model treats both minor and great disturbance in the same way, which allows the model to successfully predict the occurrence of deforestation, but it can not differentiate between high-intensity industrial clearing and low-intensity clearing within the flagged regions.

Furthermore, the extreme class imbalance that is a characteristic of deforestation data remains a challenge. Strategies such as balanced sampling, in which the majority class is under sampled, take place and loss functions such as Weighted Binary Cross Entropy and Dice Loss were used to encourage the models to focus on the minority class (Sudre et al., 2017; Yeung et al., 2022). However, these methods are probably not entirely effective. The sampling mechanism that was used during the training phase, ensures that 50% of the sampled patches contain deforestation, while the other 50% does not. The patches have a size of 64x64 pixels, and within the samples there is also an imbalance between forested pixels and deforested pixels, with the forested pixels being the majority class. Even within a sampled crop containing a deforestation event, the majority of the pixels will likely remain forested. The loss functions evaluate the performance per individual pixel, and because of the imbalance between forested and deforested pixels, the volume of forested pixels could potentially have outweighed the volume of deforested pixels. This encourages the model to predict the pixels as forested, so that it reduces its overall error (Buda et al., 2018). The consequence of this is that the models still slightly favor predicting stable forest in order to minimize loss, which limits the overall $F_{0.5}$ score, precision and recall (Johnson & Khoshgoftaar, 2019).

An additional limitation regarding the overall data quality and input features must be addressed. First of all, the target variable that is used for making the predictions relies on satellite-derived pseudo ground truth data, instead of field-verified actual ground truth. In tropical regions, the accuracy of the pseudo ground truth data is influenced by the seasonal dynamics. Since the GFW Integrated Alerts consists of GLAD and RADD alert systems, it combines both optical and radar based alerts. The optical alerts are hindered by cloud cover (Asner, 2001; Reiche et al., 2015), while the radar alerts are hindered by variations in soil moisture. High soil moisture after intense rain events can alter the radar back-scatter, which delays the detection of new deforestation events (Bouvet et al., 2018; Steele-Dunne et al., 2017). Because of this combination of limitations, and because high soil moisture and cloud

cover often occur simultaneously, the exact moment of detection of a deforestation event might be delayed because of these climatic conditions. This delay in the data might introduce noise in the chronological sequence of the target variable, because a deforestation event might go undetected for a longer period of time, meaning that an accurate prediction of the deep learning models could potentially be classified as a false positive during evaluation (Hansen et al., 2016; Pelletier et al., 2017; Reiche et al., 2021).

Additionally, the limited geographic scope of this research poses a limitation for the generalization of the results. The training and evaluation of the models was only applied on three specific countries: Gabon, Papua New Guinea and Venezuela. Even though these regions contain diverse and differing tropical forests, the model's performance was not tested in other countries or regions with active deforestation, such as the Amazon basin, the other regions of the Congo basin and South East Asia. Because of this, it remains uncertain how well the models will perform in these regions and how well they will converge in regions with different spatial features, ecological vulnerabilities and clearing methods.

Another limitation that must be addressed is the temporal scope of the dataset. The validation and test set that were used to evaluate the model's performance during and after training both consist of only a single year. This temporal restriction in time was caused by the limited availability of historical data, but it should be taken into account that this limitation introduces a temporal bias. Tropical deforestation rates are dynamic and can fluctuate from year to year because of factors such as climatic factors, economic shifts, or sudden changes in the national policies (Jayathilake et al., 2021; Pendrill et al., 2022). Because of this, there is a risk that the evaluation time-frame that was used in this thesis represents an anomalous year. If the test year was an exceptionally active deforestation year, or an unusually stable year compared to the historical data, the reported $F_{0.5}$, precision and recall scores might not represent the model's long term robustness. This temporal constraint in data causes uncertainty about how effective the trained architectures will generalize to future years.

Additionally, during the training of the models, another limitation related to dataset size and model capacity became apparent. During the training phase, the models showed very fast convergence. In all 3D deep learning models, the peak validation performance was reached within the first 15 epochs, after which the early stopping mechanism was triggered. After this peak validation performance was reached, the training loss values dropped quickly in the following epochs. This quick learning curve suggests that there might be a mismatch between the high capacity of the complex 3D architectures and the diversity or volume of the available training data. This lack of data diversity is a result of the study's constraints. The models were trained on limited geographic regions, and the temporal span of the training dataset was restricted to only two years.

As illustrated in Table 2, the 3D architectures contain roughly three times as many trainable parameters compared to the 2D baseline. The limited data volume caused quick convergence across all models, but this substantial difference in the number of parameters makes the 3D networks more susceptible to memorization, preventing them from reaching their theoretical performance of their complex spatio-temporal design (Hara et al., 2018).

Because the networks were not exposed to diverse and long term datasets, instead of slowly improving and gradually learning the complex patterns of deforestation, the deep learning models probably quickly learned the most important spatial features and began overfitting afterwards. This behavior is in line with known challenges in deep learning literature, as models require large amounts of diverse data to effectively optimize their weights and prevent

poor generalization (Thanapol et al., 2020). Next to that, the interpretability of the deep learning models is limited (Kumar et al., 2023; Luo & Xu, 2023). This quick convergence potentially prevents the models from learning the long-term changes in the input data. This could also be an explanation for the fact that the models with an extended 12-month historical context did not significantly outperform the models using a shorter 3-month context window. Furthermore, the reliance on spatial features rather than temporal sequences is in line with the findings of Cué La Rosa et al. (2025). Since both the 2D ResUNet and 3D ResUNet achieved similar scores, it could be said that explicitly modeling the temporal dimension does not result in a significant improvement over aggregating the temporal features into a single map.

A potential cause for the fact that the 3D models do not outperform the 2D model could be the geographical scope of the data. Masolele et al., 2021 showed that spatio-temporal deep learning models outperformed purely spatial models when they were trained on big, multi-continental datasets. This is also supported by Overlan et al., 2025, which used deep learning models on a larger scale region to predict near-future deforestation in the ForestCast project. On the other hand, when the models were trained on a single continent, the temporal patterns became less important and the spatial patterns had a higher influence in the final outcome of the models. In this thesis, the models were trained specifically per region. Because of this, it could be likely that the spatial patterns in these datasets were more important for the outcome of the model than the temporal patterns. This could also explain why the 2D ResUNet which is good at recognizing local patterns performs similarly to the 3D CNNs.

6.5 Broader implications and Future Research

Furthermore, it is important to place these results in the context of other remote sensing literature. A review by Md Jelas et al., 2024 confirms that deep learning models, and U-net variants in particular, are very effective models for deforestation applications. However, a general observation across the current literature available is that the majority of the papers focus on the detection of past or current deforestation, instead of near-future prediction (Bai et al., 2023; Haq et al., 2024; Jiang et al., 2022). Predicting future deforestation events is a fundamentally different and more complex task. Detection relies mainly on identifying visible changes in spectral signatures, while prediction requires the models to anticipate environmental, geographic, and human drivers before the deforestation event occurs.

The findings of this thesis are useful for the application of the Forest Foresight initiative and for the theory. The results indicate that organizations do not necessarily need to invest time and money in regionally training 3D spatio-temporal models for general predicting of deforestation. The 2D ResUNet that utilizes the aggregated temporal dimension is already a robust model that is more computationally efficient that can be used for the same task with similar performance. Theoretically, the results show a prospect for future research. Because the explicit temporal modeling did not significantly improve model performance on the regional scale, future studies should investigate whether training a 3D architecture on a pan-tropical dataset extracts the theoretical improved performance of the spatio-temporal convolutional models. Exposing the model to a bigger volume and variety of deforestation events across continents might provide sufficient data to effectively let the model learn the temporal dependencies (Kumar et al., 2023), which could potentially lead to a better early warning deforestation prediction.

7 Conclusions

The goal of this thesis was to evaluate the added value of explicitly modeling the temporal dependencies of deforestation, with the goal of improving the spatio-temporal deep learning models that are used for predicting deforestation. The most important finding is that 3D spatio-temporal models do not provide a universal, significant improvement in relation to models that aggregate the historical context into a single feature map. When comparing different deep learning architectures, it turned out that the convolutional models (2D ResUNet, 3D ResUNet and ConvLSTM3D) resulted in comparable results, and they significantly outperformed the attention-based ViViT model. The 2D ResUNet model resulted in the highest average $F_{0.5}$ score of 0.6526, which implied that a high spatial awareness and inductive bias is important for this prediction task.

Regarding the optimal temporal window, it turned out that a short historical context of 3 months was sufficient to capture the relevant dynamics of deforestation progression. The 3-month context model achieved a similar average $F_{0.5}$ score as the 6-month context and 12-month context, which implied that adding longer historical sequences does not introduce significant predictive values to the models, and could potentially even introduce noise in the learning trajectory of the models.

During the evaluation of the model performance on different prediction horizons, it was noted that the models perform significantly better over longer prediction horizons. A prediction horizon of 12-months resulted in the highest average $F_{0.5}$ score of 0.7222, while the model with a 3-month prediction horizon only achieved an average $F_{0.5}$ score of 0.5068. This difference in performance is the result of the nature of the target variable. Over the span of a longer prediction horizon, there is a greater accumulation of deforestation events and forest disturbances. This causes there to be a greater number of deforested pixels, which makes it easier for the model to classify these pixels.

Training and evaluating the models on different regions highlighted the strong geographical differences. Notably, in Gabon, the 3D ResUNet identified a significantly larger portion of new deforestation events when compared to the 2D ResUNet. However, in Papua New Guinea this was exactly the other way around. Here the 2D ResUNet outperformed the 3D ResUNet on the specific task of predicting new deforestation events. In Venezuela, both model architectures performed poorly, with $F_{0.5}$ scores between 0.43 and 0.53. This was likely caused by climatic anomalies in the test set, such as big forest fires in 2024. The global performance over all pixels for all models was similar between the 2D and 3D architectures, where no significant differences was between them.

Despite the insights, there are some limitations and gaps in this research. The extreme class imbalance of target variable, and the simplification that exists due to the binarization of the ground truth limit the performance of the models. Additionally, the limited geographic extent and the short time range over which the models were trained, validated and tested limited the optimal utilization of the more complex, heavier 3D models. Future research could improve on this thesis by extending the temporal and spatial scope, or by switching from a classification to a regression task, so that binarization of the ground truth is not required.

8 Recommendations

Based on the identified limitations in this research, a number of recommendations for future research can be made to further optimize the prediction of tropical deforestation using spatio-temporal deep learning models.

First of all, the research showed that the complex 3D models, like the 3D ResUNet, converged very quickly after initializing the training. The best performing model on the validation set was usually found within the first 15 epochs, after which the early stopping mechanism triggered. This points to possible overfitting in further epochs, which could be caused by a lack of sufficient spatial and temporal diversity in the regional datasets. Since these datasets for training consisted of only two years, in relatively small regions, this is a highly probable cause for the quick convergence.

To fully utilize the theoretical capacity of the spatio-temporal convolutional networks, it is recommended to train these architectures on bigger datasets, for example, a pan-tropic dataset that covers multiple countries and continents simultaneously. Following the methodology of a large scale benchmark such as the ForestCast project (Overlan et al., 2025) could expose the models to a greater number and more diverse set of deforestation events which can allow them to learn the spatial and temporal dependencies more effectively. Additionally, it is recommended to use a longer time period for training, validating, and testing, to reduce the effect that seasonal or yearly variations have on the models. By combining multiple regions over a longer period of time, the models are exposed to a significantly greater number of deforestation events and dynamics. This will potentially allow the models to learn the complex temporal dependencies effectively without overfitting on local spatial patterns.

Additionally, the current methodology reduces the very detailed ground truth variable to a binary dataset for classification. Because of this, the model treats a pixel with only 1 deforested sub-pixel in the exact same way as a pixel that has been completely cleared. Even though this allows the model to predict the occurrence of deforestation events relatively successfully, it is incapable of making a differentiation between low-intensity forest disturbances and large-scale, industrial-level forest clearing.

To combat this limitation, it is recommended to develop a new model architecture that switches from a classification task, like in this thesis, to a regression task. This way, the ground truth variable does not have to be binarized, and the original ground truth pixel values (ranging from 0 to 1600 deforested sub-pixels) can be maintained. The models will need to be adjusted with specific regression loss functions, such as Mean Squared Error (MSE), to ensure compatibility with this new task. In theory, this could lead to more detailed results in which the actual intensity of future deforestation events can be predicted, which could significantly increase the practical value of the models for governments and enforcement monitoring agencies.

Lastly, the target variable that is used for making predictions in this thesis is based on satellite-derived pseudo ground truth data, specifically the GFW Integrated alerts. This data is susceptible to climatic conditions such as cloud cover or high soil moisture that alters radar backscatter. This can lead to delayed detection of deforestation events, which introduces chronological noise in the dataset. An accurate early prediction made by the model might be unfairly classified as a false positive if there is a delay in the detection of the GFW Integrated Alerts.

Future research could attempt to mitigate this limitation. This could for example be achieved

by incorporating the confidence levels that are available in the GFW alerts into the model evaluation metrics. pixels with low confidence could be assigned lower weights than pixels with high confidence for example. By accounting for this detection delay and confidence, the true predictive accuracy of the models for predicting near-future deforestation could be evaluated more reliably.

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10 Statement on the use of Generative AI

During the development of this thesis, generative AI was used in accordance with the university guidelines. The AI was used exclusively as a sparring partner and a feedback tool, and under no circumstances was it used to generate new information or content for this thesis. The application of AI was strictly limited to programming support and writing feedback. The conceptual design and the training and evaluation pipeline used in this research are developed from my own choices. Generative AI was only used to support the technical implementation of this pipeline by assisting with coding syntax and troubleshooting errors. All text and academic reasoning in this report was written independently. Generative AI was used only as a proofreading and editing tool to check grammar and spelling. Additionally, it was used to receive feedback on the logical reasoning and the stylistic flow of the writing.

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Annex: Research Data Archive

This annex outlines the contents of the research data archive (ZIP file) submitted alongside this thesis. The archive is structured according to the following elements:

- **Documentation:** Documentation detailing the contents and locations within the archive, including the complete folder structure.
- **Report:** The final thesis report (provided in Word and PDF formats).
- **Presentations:** The slides for both the midterm and final presentations.
- **Datasets:** All raw, processed, and created datasets used during this research.
- **Figures and Maps:** Original files for figures, maps, and tables.
- **Scripts and Code:** All scripts and code for data processing, model training, and evaluation.